Nombre: Exámen

Tema:

Día | Mes | Año | Folio

1)

$E_c = ¿?$

$B = 5.5 [mT] \rightarrow 0.0055[T]$

$r = 6 [mm] \rightarrow 0.006[m]$

$q = 1.6022 \times 10^{-19} [C]$

$m = 9.1094 \times 10^{-31} [kg]$

$$E_c^{¿?} = \frac{1}{2} m \cdot V^{2\,¿?}$$

$$\frac{q}{m} = \frac{V^{¿?}}{B \cdot r}$$

$$V = \frac{B \cdot q \cdot r}{m}$$

$$V = \frac{(0.0055[T])(1.6022 \times 10^{-19}[C])(0.006[m])}{(9.1094 \times 10^{-31}[kg])}$$

$$V = 5804180.298 \left[\frac{m}{s}\right]$$

$$E_c = \frac{1}{2}(9.1094 \times 10^{-31})\left(5804180.298\left[\frac{m}{s}\right]\right)^2$$

$$E_c = 1.53441051 \times 10^{-17} [J]$$

Nombre: Día | Mes | Año | Folio

Tema: $\varepsilon_{f_1} = (4.3183\times10^{-19})(8.43\times10^{23})$

2) $h = 6.62607\times10^{-34}$
$c = 2.9979\times10^{8}$
ε_{f_1} = Fotones totales $= 8.43\times10^{23}$
$\lambda = 460\,[nm] = 4.6\times10^{-7}$

$$\varepsilon_f = \frac{(6.62607\times10^{-34})(2.9979\times10^{8})}{4.6\times10^{-7}}$$

$\varepsilon_{f_1} = 364032.69\,[J]$

Δ_e Total $= 364032.69 - 253717.71$

$\Delta_e = 110314.98\,[J]$

$\Delta_{e\,Total} = \varepsilon_{f_1} - \varepsilon_{f_2}$

$$\varepsilon_f = \frac{h\cdot c}{\lambda}$$

ε_{f_2} = Fotones totales $= 8.43\times10^{23}$

$$\varepsilon_{f_2} = \frac{(6.62607\times10^{-34})(2.9979\times10^{8})}{6.6\times10^{-7}}$$

$\varepsilon_{f_2} = (3.0097\times10^{-19})(8.43\times10^{23})$

$\varepsilon_{f_2} = 253717.71\,[J]$

3) $\lambda = 50.69\,[nm] \rightarrow 5.069\times10^{-8}\,[m]$
$W_0 = 21.29\,[eV] \rightarrow 3.411034\times10^{-18}\,[J]$
$h = 6.62607\times10^{-34}\,[J\cdot s]$
$c = 2.9979\times10^{8}\,[m\cdot s^{-1}]$
$v = ¿?$

$$\varepsilon_f = \frac{h\cdot c}{\lambda}$$

$\varepsilon_f = W_0 + \varepsilon_c$

$\varepsilon_c = \frac{1}{2} m\cdot v^2$

$$v = \sqrt{\frac{\varepsilon_c\cdot 2}{m}}$$

$$\varepsilon_f = \frac{(6.62607\times10^{-34})(2.9979\times10^{8})}{5.069\times10^{-8}}$$

$\varepsilon_f = 3.918\times10^{-18}\,[eV]$

$\varepsilon_c = \varepsilon_f - W_0$
$\varepsilon_c = 3.918\times10^{-18} - 3.411034\times10^{-18}$
$\varepsilon_c = 5.0744\times10^{-19}\,[\frac{m}{s}]$

Nombre:
Tema:

Día	Mes	Año	Folio

$$v = \sqrt{\frac{(5.0774 \times 10^{-18})2}{9.1093 \times 10^{-31}}}$$

$v = 1055461.983 [m/s]$

4)

$mv = 8.43657 \times 10^{-34}$

$r = 4.3296 \times 10^{-11}$

$\lambda = ¿?$

$$n = \frac{(m \cdot v \cdot r)(2\pi)}{h}$$

$$\lambda_c = \frac{2\pi \cdot r}{n}$$

$$n = \frac{(8.43657 \times 10^{-34})(4.3246 \times 10^{-11})(2\pi)}{6.62607 \times 10^{-34}}$$

$n = 346.367821901 \times 10^{-80}$

$$\lambda_e = \frac{2\pi(4.3296 \times 10^{-11}}{346.367821901 \times 10^{-80}}$$

$\lambda_c = 6.86324106 \times 10^{64} [m]$

Nombre:

Tema:

Día | Mes | Año | Folio

5)

$E_c = 0.32\,[eV] \rightarrow 5.1269 \times 10^{-20}$

$h = \frac{c}{\lambda}$

$h = 6.62607 \times 10^{-34}\,[J \cdot s]$

$\lambda = h \frac{c}{E_c}$

$c = 2.9999 \times 10^{8}\,[m \cdot s^{-1}]$

$\lambda = ¿?$

$$\lambda = 6.62607 \times 10^{-34}\,\frac{2.9979 \times 10^{8}}{5.1269 \times 10^{-20}}$$

$$\lambda = 3.8745 \times 10^{-6}\,[m]$$

6) Nb^{2+}

$41e^- + 2e^- = 39e^-$

$1s^2$: $\uparrow\downarrow$ (0)

$2s^2$: $\uparrow\downarrow$ (0)

$2p^6$: $\uparrow\downarrow$ (−1), $\uparrow\downarrow$ (0), $\uparrow\downarrow$ (+1)

$3s^2$: $\uparrow\downarrow$ (0)

$3p^6$: $\uparrow\downarrow$ (−1), $\uparrow\downarrow$ (0), $\uparrow\downarrow$ (+1)

$4s^2$: $\uparrow\downarrow$ (0)

$3d^{10}$ (30): $\uparrow\downarrow$ (−2), $\uparrow\downarrow$ (−1), $\uparrow\downarrow$ (0), $\uparrow\downarrow$ (+1), $\uparrow\downarrow$ (+2)

$4p^6$: $\uparrow\downarrow$ (−1), $\uparrow\downarrow$ (0), $\uparrow\downarrow$ (+1)

$5s^2$: $\uparrow\downarrow$ (0)

$4d^{10}$: $\uparrow$ (−2), $\uparrow$ (−1), $\uparrow$ (0), — (+1), — (+2)

a) $n = 5$, se cumple en $5s^2$, el cual cuenta con dos e^-

b) $l = 3$, no se cumple en ningun electión

c) $m = -2$, se cumple en $3d^{10}$ y $4d^{10}$, en total hay 3 electrónes

Nombre:
Tema:

Día | Mes | Año | Folio

d) $l = 0$ y $m = -2$, No se cumple en ninguno

7) Ti^{3+} $Z = 22e^- + 3e^- = 19e^-$

$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$

$1s^2$: ↑↓ (0)

$2s^2$: ↑↓ (0)

$2p^6$: ↑↓ ↑↓ ↑↓ (−1, 0, +1)

$3s^2$: ↑↓ (0)

$3p^6$: ↑↓ ↑↓ ↑↓ (−1, 0, +1)

$4s^2$: ↑↓ (0)

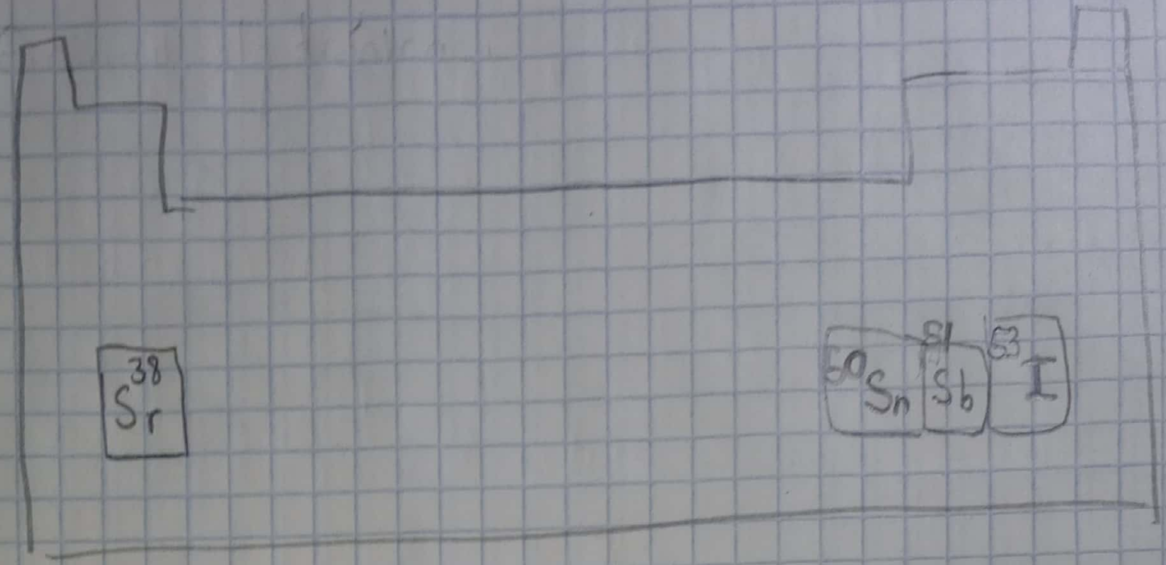
$3d^{10}$: ↑ ↑ _ _ _ (−2, −1, 0, +1, +2)

$n = 4$
$l = 0$
$m = 0$
$s = +1/2$

Además, es un ión no magnético

Nombre: | Día | Mes | Año | Folio
Tema:

8)



Afinidad electrónica ↑ Aumenta →

Sr, Sn, Sb, I

Primera energía de Ionización ↑ Aumenta →

Sr, Sn, Sb, I

Electronegatividad ↑ Aumenta →

Sr, Sn, Sb, I

Radio atómico ↓ Aumenta ←

I, Sb, Sn, Sr